Vishal S I

B.Tech CS Graduate | Full-Stack Engineer

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SUMMARY

B.Tech CS graduate from VIT with 4 production-level internships spanning computer vision, NLP, full-stack engineering, and backend ML. Built and deployed REST APIs, NLP classification pipelines, and a cashierless retail CV system used in real-world settings. Strong at identifying the gap between benchmark performance and production reliability. Seeking full-time SDE or ML Engineer roles with industry experience.

EXPERIENCE

AI / ML Intern — Computer Vision & Model Evaluation
Digit7 India Pvt. Ltd. · India
Jan 2026 – May 2026

- Evaluated annotated training data and model outputs for a cashierless retail CV system using shelf-mounted cameras and customer movement tracking.
- **Discovered systematic annotation failures** in occlusion scenarios that caused confident but incorrect predictions — invisible to aggregate metrics; designed targeted evaluation protocols to surface this failure class.
- Developed deep understanding of error propagation across multi-model vision pipelines and the divergence between aggregate metric performance and real-world failure modes.

Full Stack Development Intern
SST Cloud Solutions Pvt. Ltd(Client: National Rugby League, Australia) · Coimbatore
May 2025 – Jul 2025

- Built scalable REST APIs in Node.js and Express.js; integrated React.js frontend components with secured backend services for a live production system serving real users.
- Resolved Level-2 engineering issues in production by diagnosing root causes via log analysis and implementing sustainable fixes under active user load.
- Identified API response inefficiencies and proposed targeted optimisations that measurably reduced latency under load.

Academic Intern — Machine Learning
NUS School of Computing · Singapore
Dec 2024 – Jan 2025

- Built Ridge, Lasso, and gradient-boosted tree regression models for profit margin prediction across ~3,000 retail transactions.
- **Key finding:** 3 engineered ratio features accounted for >60% of variance explained; 12 additional features added noise — derived from statistical assumption analysis, not metric optimisation alone.
- Completed Big Data Analytics using Deep Learning certification, credit-transferred to VIT undergraduate record.

Frontend Intern
GreenOrange Information Technology Pvt. Ltd. · Coimbatore
May 2024 – Jul 2024

EXPERTISE

- **ML / AI:**
  Machine Learning, Deep Learning, NLP, Computer Vision, TF-IDF, DistilBERT, MiniLM, SVM, Regression, Evaluation Design
- **Full-Stack Development:**
  FastAPI, Express.js, Node.js, React.js, Next.js, Flutter, REST APIs, Firebase
- **Languages:**
  Python, JavaScript, Java, C++, Dart
- **Cloud & Databases:**
  AWS, Amazon SageMaker, MongoDB, Supabase, SQL
- **Production Engineering:**
  Docker, Git, API design, log-driven debugging, latency optimisation
- **Evaluation & Research:**
  Benchmark vs. real-world failure analysis, annotation quality, error propagation in multi-model pipelines

CERTIFICATIONS

- Big Data Analytics using Deep Learning
  NUS School of Computing (Corporate Gurukul)
  Dec 2024 – Jan 2025
- Deep Learning with Amazon SageMaker
  AWS Certification
  Dec 2024

EDUCATION

- **B.Tech in Computer Science & Engineering**
  Vellore Institute of Technology (VIT)
  2022 – 2026
  CGPA: 7.57 · IELTS: 7.0 (C1)
- **Class XII**
  82% · Coimbatore, India

TECHNICAL SKILLS

- ML / NLP / Computer Vision
- REST API Design & Deployment
- Full-Stack Web & Mobile Dev
- Production Debugging & Optimisation
- Model Evaluation & Failure Analysis
- RAG Systems & LLM Integration
- Cloud Deployment (AWS, SageMaker)
- Data Engineering & Feature Design
- Agile Collaboration & Git Workflows
- Statistical Modelling & Regression

- Built mobile UI screens and components using Flutter/Dart, integrating with backend REST APIs to deliver a complete end-to-end user experience.
- Collaborated on FastAPI-based backend services that powered the Flutter application, gaining exposure to full-stack data flow and API contract design.
- Achieved 87% model accuracy in deployed ML-backed features; developed practical understanding of the gap between benchmark accuracy and production reliability.

PROJECTS

AI Customer Support Ticket Classifier
Python · FastAPI · Scikit-learn · DistilBERT · Sentence-BERT

- Built end-to-end NLP pipeline: text preprocessing, TF-IDF vectorisation, and benchmarking of LR, SVM, Naïve Bayes, DistilBERT, and Sentence-BERT across 555 custom-curated synthetic tickets.
- 96.5% accuracy, Macro F1: 0.96 — TF-IDF outperformed DistilBERT on short structured text; Naïve Bayes outperformed LR on low-frequency categories (regularisation effect under data scarcity).
- Deployed via FastAPI with confidence-based human-in-the-loop routing; auto-flags predictions below 0.45 confidence threshold.

RankGym — Gamified iOS Fitness App
Next.js 15 · TypeScript · Capacitor · Apple HealthKit · Framer Motion · Tailwind CSS

- Built a full iOS fitness app (94.4% TypeScript) that turns workout tracking into an RPG — users progress through E-to-S class ranks by completing daily quests, logging lifts, and levelling up stats.
- Integrated Apple HealthKit via Capacitor to sync real-time step count, active calories, and sleep duration; all user data stored locally with no external server dependency.
- Engineered a stat calibration system that computes a user's starting rank from real-world benchmarks and renders a dynamic radar chart reflecting live fitness progress.
- Implemented a quest engine with daily auto-generated tasks, rare Sudden Quest events, XP tracking, and streak system to reinforce consistent training habits.

MaligaiKadai — Retail Management System
React.js · FastAPI · MongoDB

- Designed normalised schemas for products, suppliers, and transactions; automated invoice generation and real-time inventory reconciliation via REST APIs.

Vehicle Tracking Dashboard
React.js · Node.js · REST APIs

- Integrated real-time backend APIs with interactive frontend visualisations for logistics tracking; modular pipeline design under latency constraints.